Requirement Analysis Phase

Technology Stack(Architecture & Stack)

1. Technology Stack

- **Core Platform Engine:** ServiceNow Enterprise Cloud Platform (Global Application Scope).
- **Application Lifecycle Management:** ServiceNow Studio & Global Update Sets (ACL Project [Global]).
- **Scripting Engine:** Server-Side JavaScript (ECMAScript 2021 / ES12 standard mode).
- **Security Framework:** High Security Settings & security_admin Privileged Elevation Model.
- **Database Management System:** Relational Database Engine with Auto-Numbering (getNextObjNumberPadded()).

2. Architectural Layers & Data Governance

Presentation & UI Layer

- User Interfaces: ServiceNow Next Experience UI, Application Navigator, Form & List Views (u_institution_details.list).
- Dynamic UI Controls: Automatic conditional rendering of platform actions (e.g., hiding/showing the New button via Create ACLs).

Application Security Engine Layer

- Access Control List (ACL) Engine: Evaluates security rules sequentially (Role Check -> Data Condition -> Advanced Script).
- Role Validation: Verifies operational roles (bb1 for Read, bb2 for Create, bb3 for Write, bb4 for Delete).
- Attribute Filtering: Evaluates field conditions (Branch == EEE) at database runtime.
- Server Script Execution: Executes gs.hasRole() functions dynamically to enforce row-level data isolation.

3. Data Schema & Dictionary Definition

Field Label	Element Name	Data Type	Reference / Configuration Details
Student Roll Number	u_student_roll_number	String	Auto-Number (Padded sequence generator)
Student Name	u_student_name	Reference	Points to sys_user table
Faculty Name	u_faculty_name	Reference	Points to sys_user table
Branch	u_branch	Choice	Choice Options: ECE, EEE, CSE
Email	u_email	String	Standard contact email string
Phone Number	u_phone_number	String	Standard contact phone string
Description	u_description	Multi String	Multi-line detail description field

4. End-to-End Execution Flow Architecture



5. Operational ACL Matrix

ACL Operation	Target Element	Required Role	Evaluation Logic & Enforcement Criteria
Read	u_institution_details	bb1	Evaluates Branch == EEE + Script Check (gs.hasRole('bb1'))
Create	u_institution_details	bb2	Validates role bb2 presence; enables New record creation button
Write	u_institution_details	bb3	Validates role bb3 presence; grants field edit rights
Delete	u_institution_details	bb4	Validates role bb4 presence; authorizes single/bulk deletion